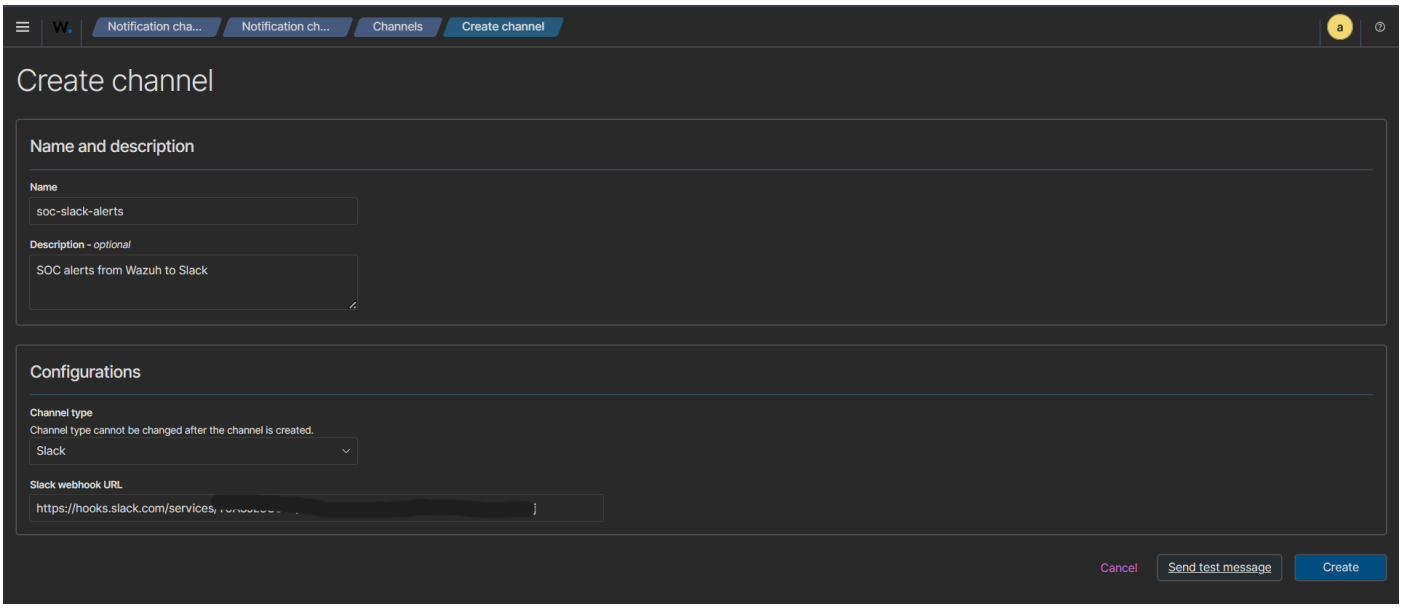
SSH Brute Force Detection and Alerting using Wazuh, Slack, and Kali Linux

➢ Project Objective

The objective of this project is to simulate an SSH brute force attack in a controlled lab environment and configure Wazuh to detect the attack and send real-time alerts to a Slack channel. This project demonstrates end-to-end SOC alerting, from attack simulation to analyst notification.

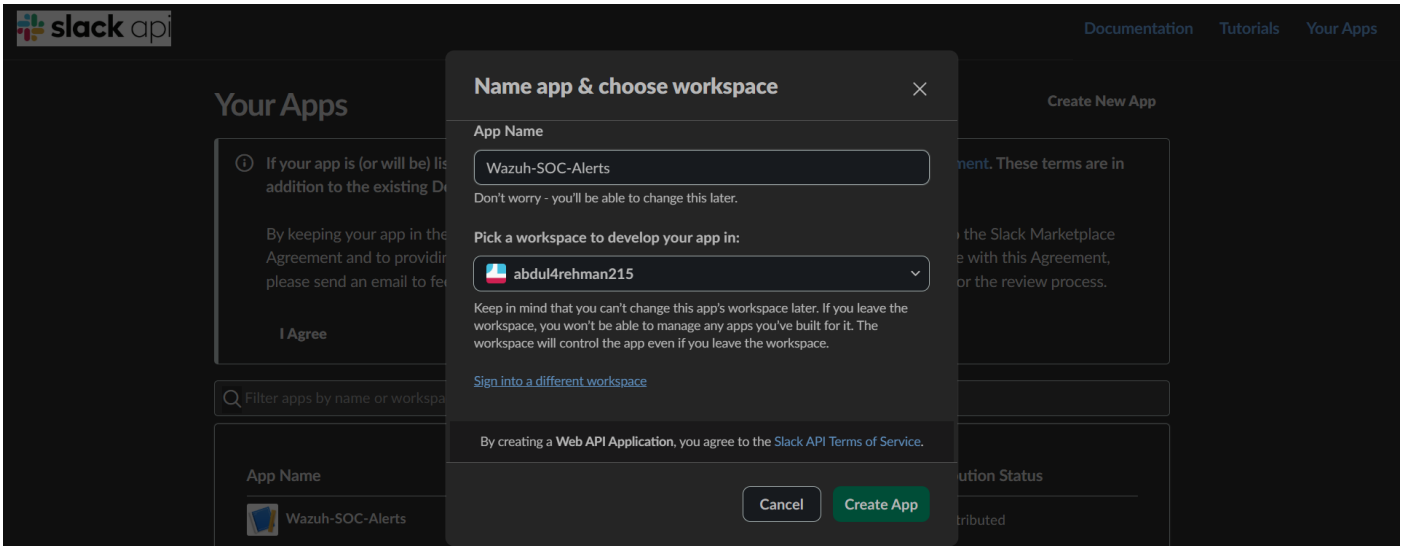
Step 1: Creating Alert Notification Channel in Wazuh

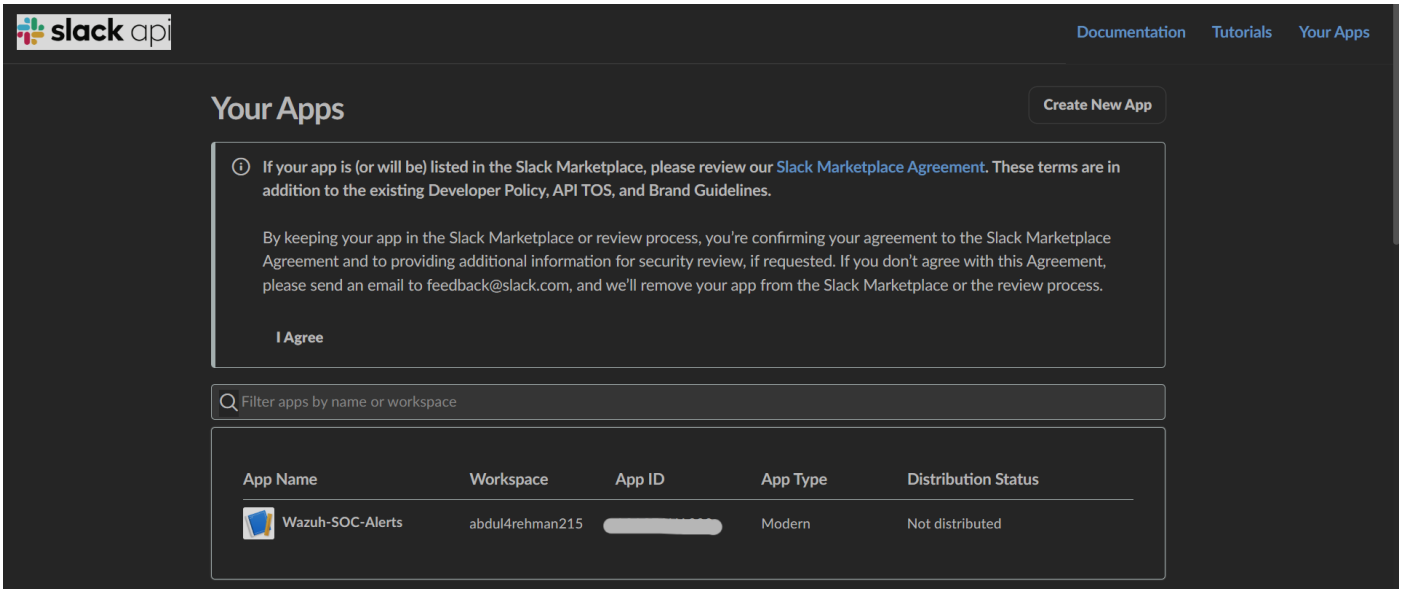
To enable alert delivery, the first step was to create a dedicated notification channel in Wazuh. This channel is responsible for sending security alerts from Wazuh to an external communication platform.



Step 2: Setting Up Slack App for SOC Alerts

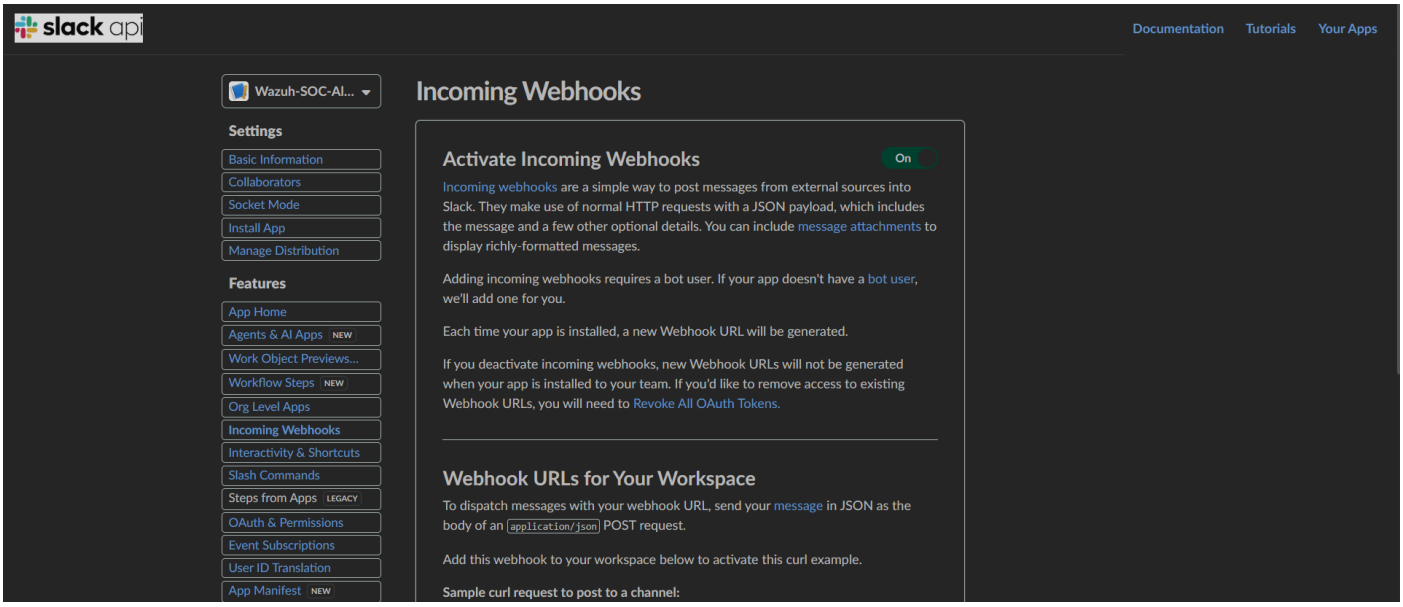
A new Slack application was created specifically for SOC alerting purposes. This app is used to receive alerts from Wazuh and post them into a Slack channel in real time.





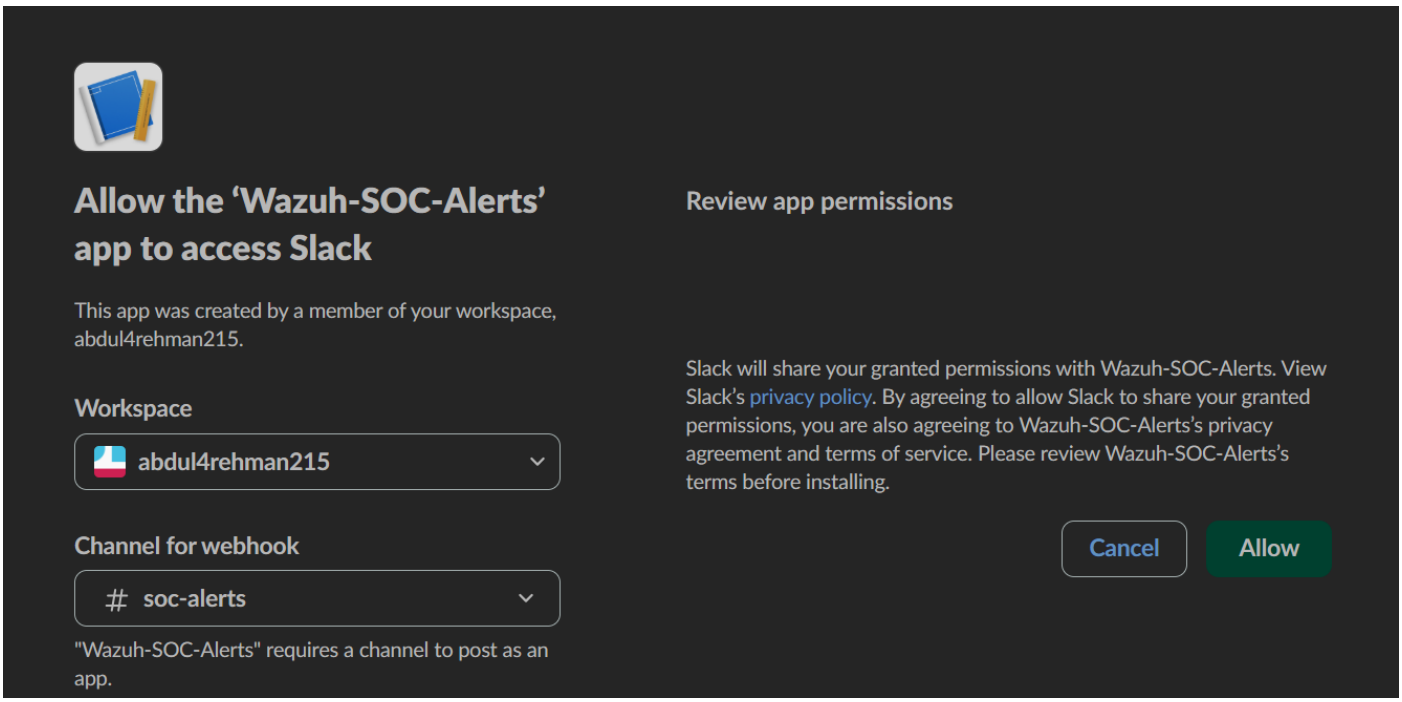
3: Enabling Incoming Webhooks in Slack

Incoming Webhooks were enabled in the Slack application to allow Wazuh to send alert messages. A webhook URL was generated, which acts as the endpoint for receiving alerts.



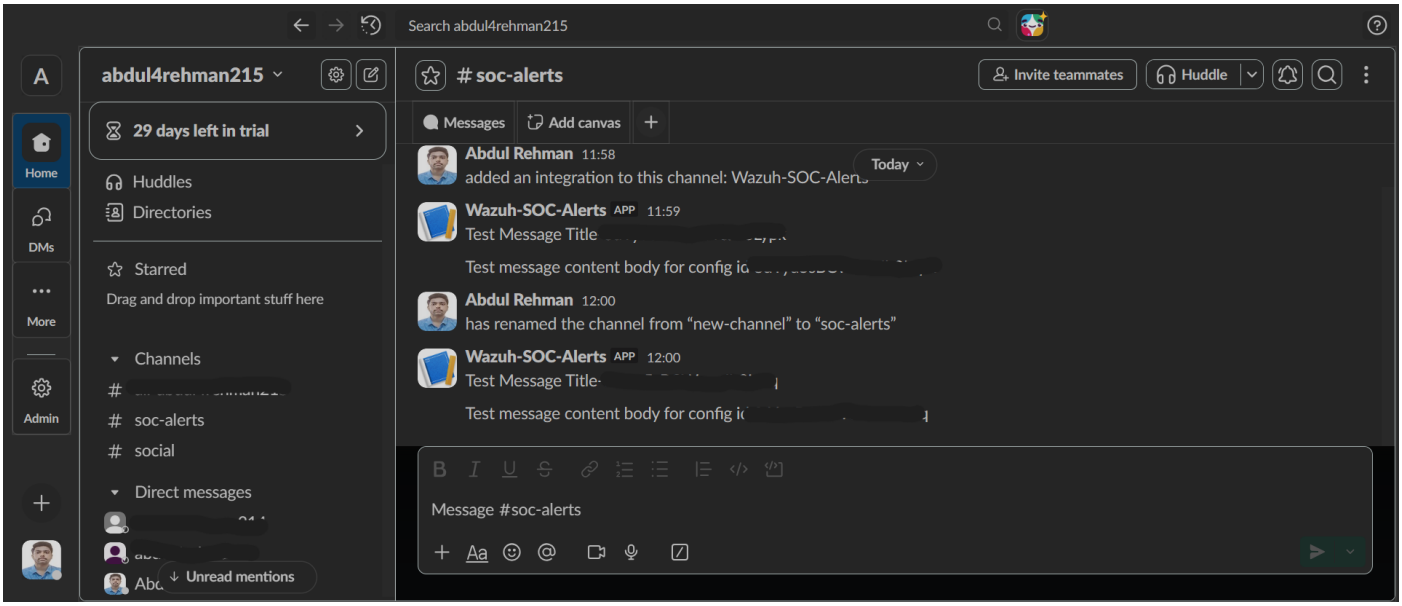
Step 4: Granting Slack Permissions and Selecting Alert Channel

The Slack app was authorized to access the workspace and post messages to the dedicated SOC alerts channel. This ensures alerts are visible to analysts immediately.



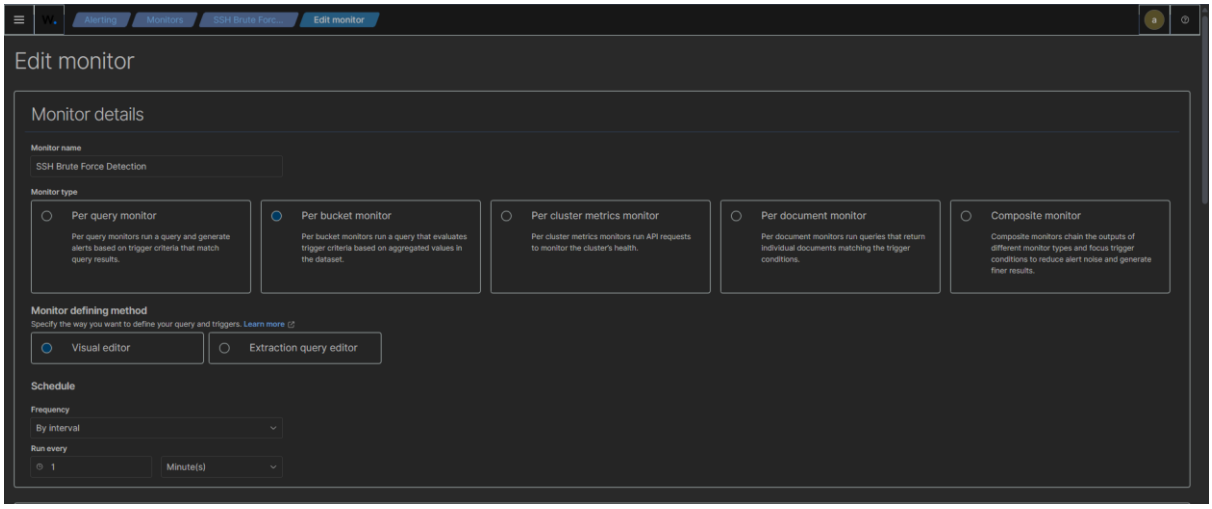
Step 5: Verifying Slack Alert Channel Connectivity

Test messages were successfully received in the Slack channel, confirming that the integration between Wazuh and Slack was working correctly.



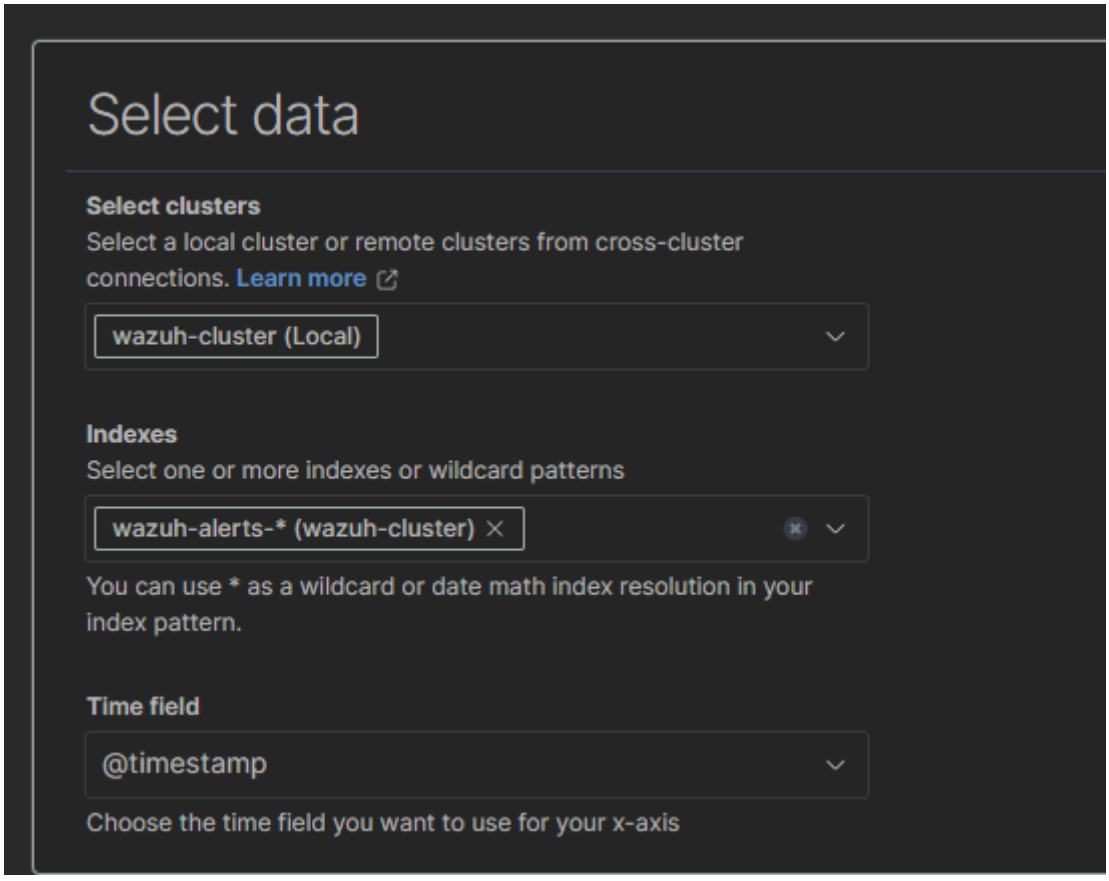
Step 6: Creating SSH Brute Force Detection Monitor in Wazuh

A detection monitor was created in Wazuh to identify SSH brute force activity. The monitor was configured to analyze SSH authentication failure events collected from the client machine.



Step 7: Selecting Wazuh Alert Data Source

The monitor was configured to read data from Wazuh alert indexes and use the timestamp field for time-based analysis.



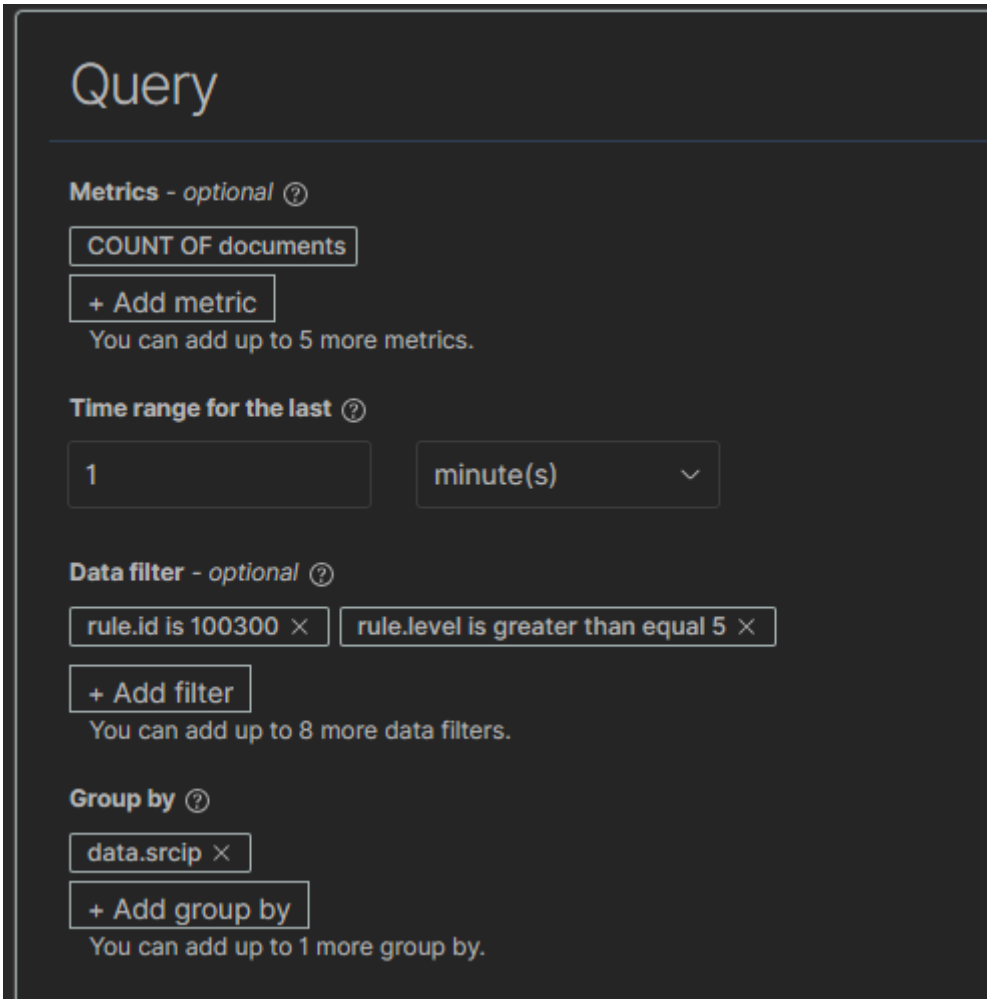
Step 8: Defining Detection Logic for SSH Brute Force

The detection logic was configured to:

- Count SSH authentication failure events
- Filter by SSH brute force rule ID
- Consider only higher severity events
- Detect multiple failures within a short time window

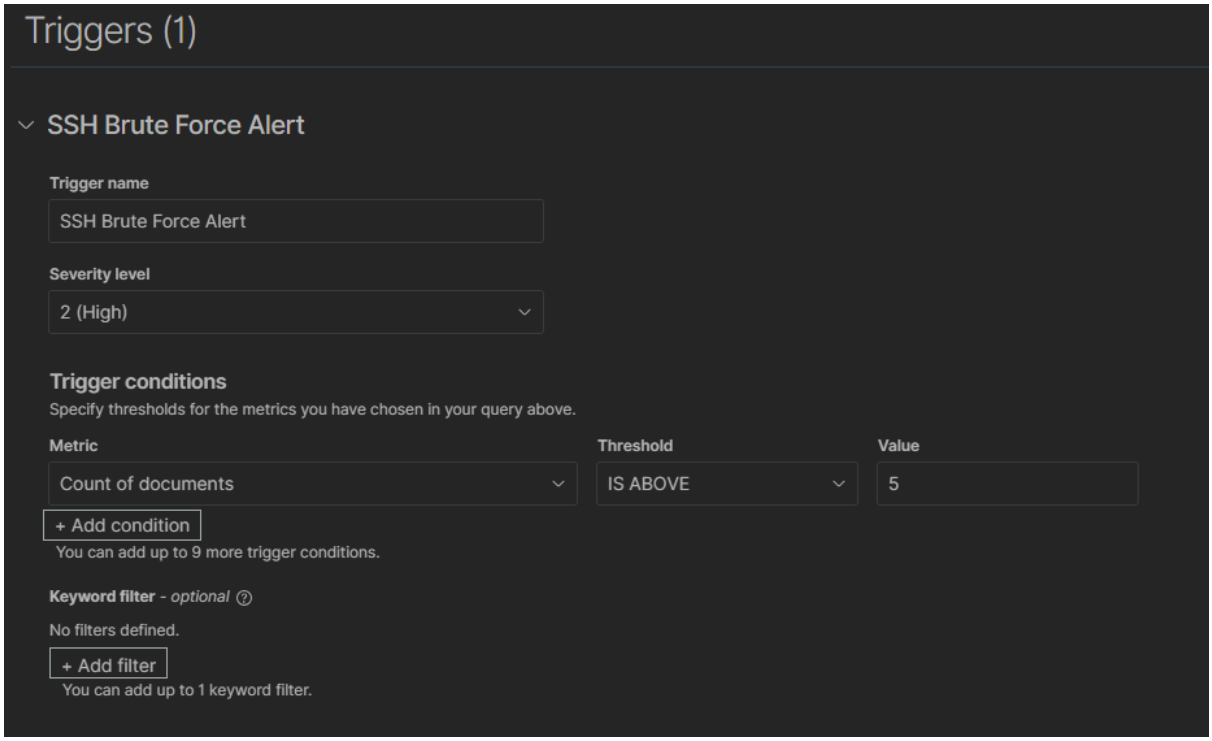
Step 9: Grouping Events by Attacker Source IP

Events were grouped by source IP address to correctly identify brute force attempts coming from the same attacker instead of treating each event individually.



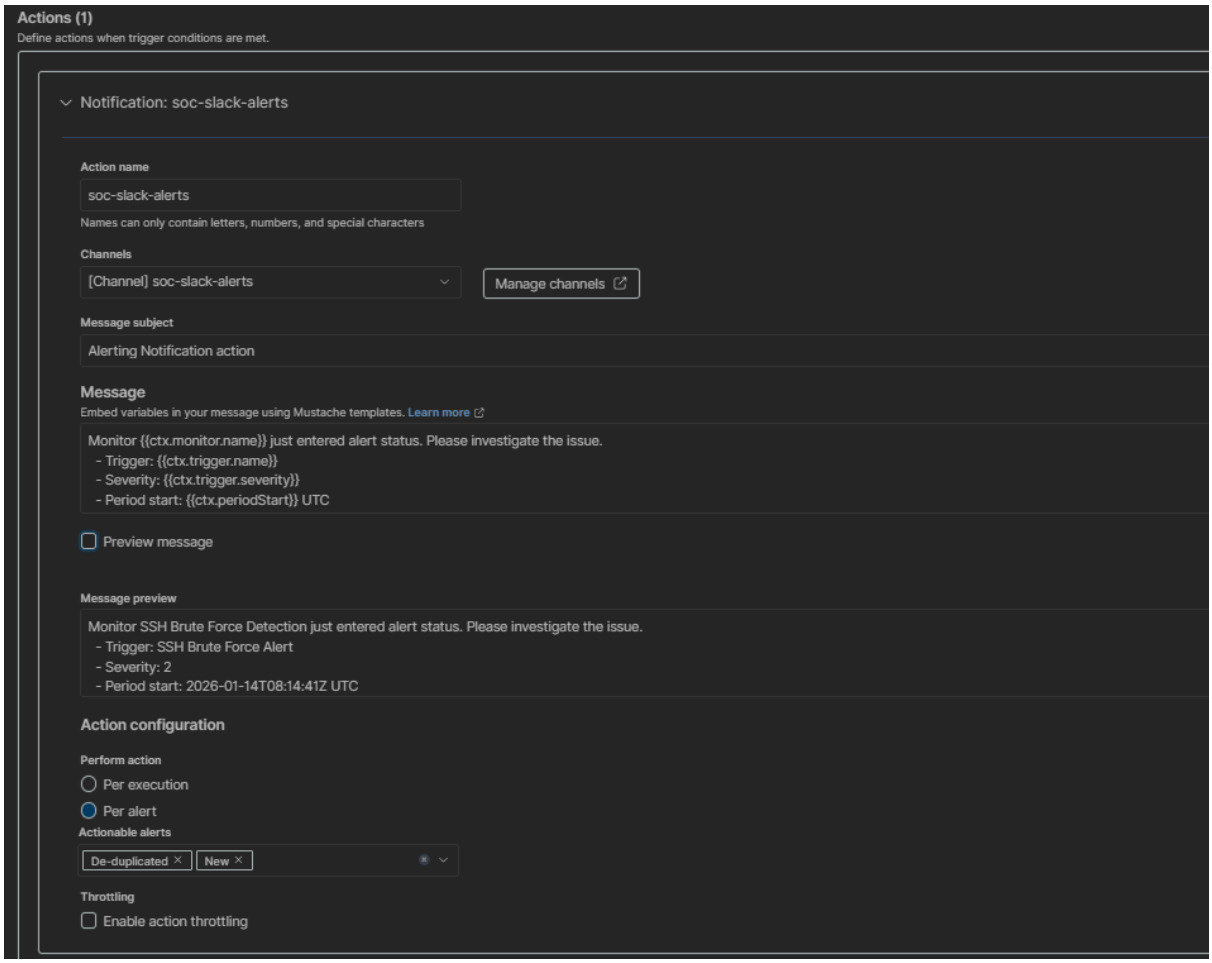
Step 10: Configuring Alert Trigger Threshold

An alert trigger was configured to fire when the number of SSH authentication failures exceeded a defined threshold within one minute. This helps reduce false positives while still detecting real attacks.



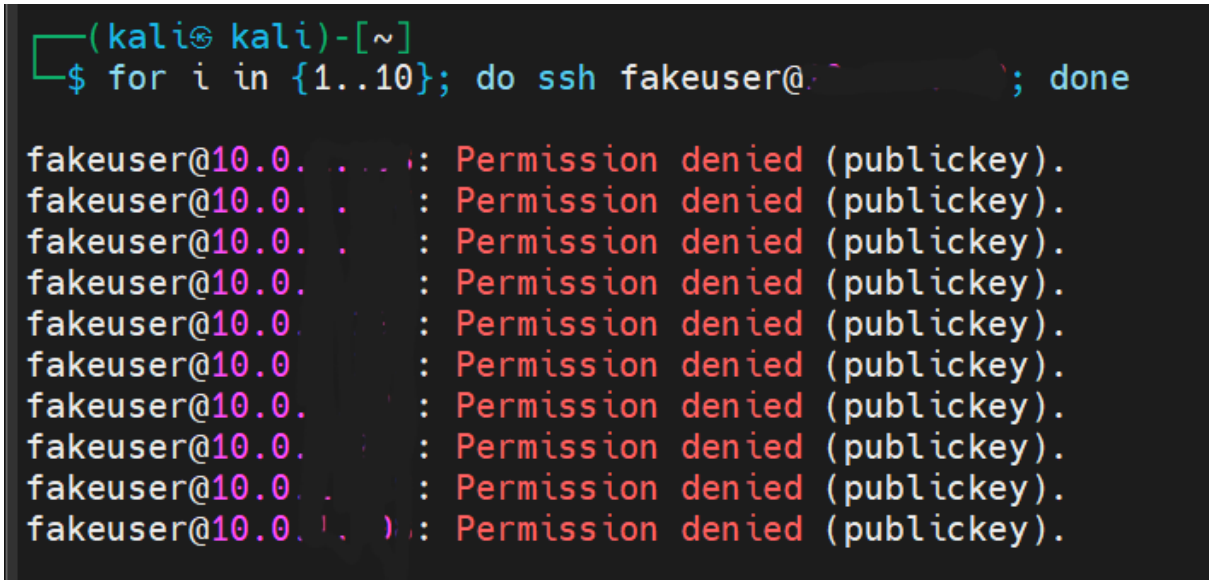
Step 11: Linking Slack Notification Action to Alert

The previously created Slack notification channel was attached to the alert so that whenever the brute force condition is met, an alert is sent automatically to Slack.



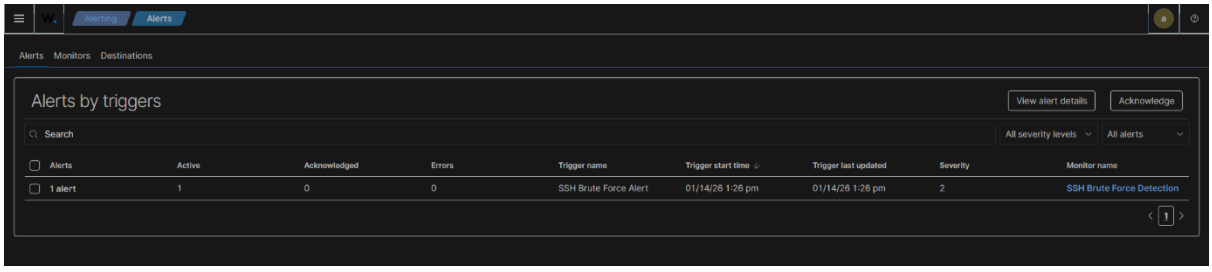
Step 12: Simulating SSH Brute Force Attack from Kali Linux

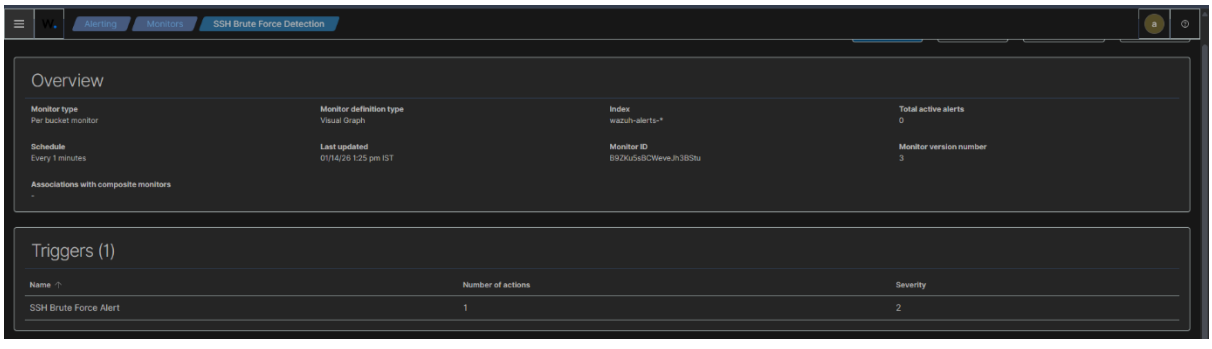
An SSH brute force attack was simulated from the Kali Linux attacker machine by repeatedly attempting SSH logins using invalid usernames against the client machine.

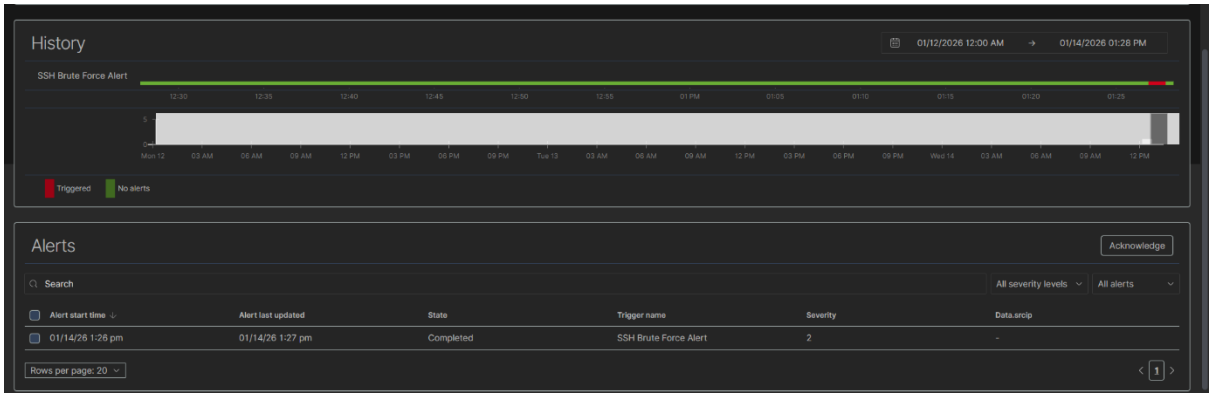


Step 13: Detection and Alert Generation in Wazuh

Wazuh successfully detected the repeated SSH authentication failures, correlated them as a brute force attack, and triggered the alert based on the configured logic.

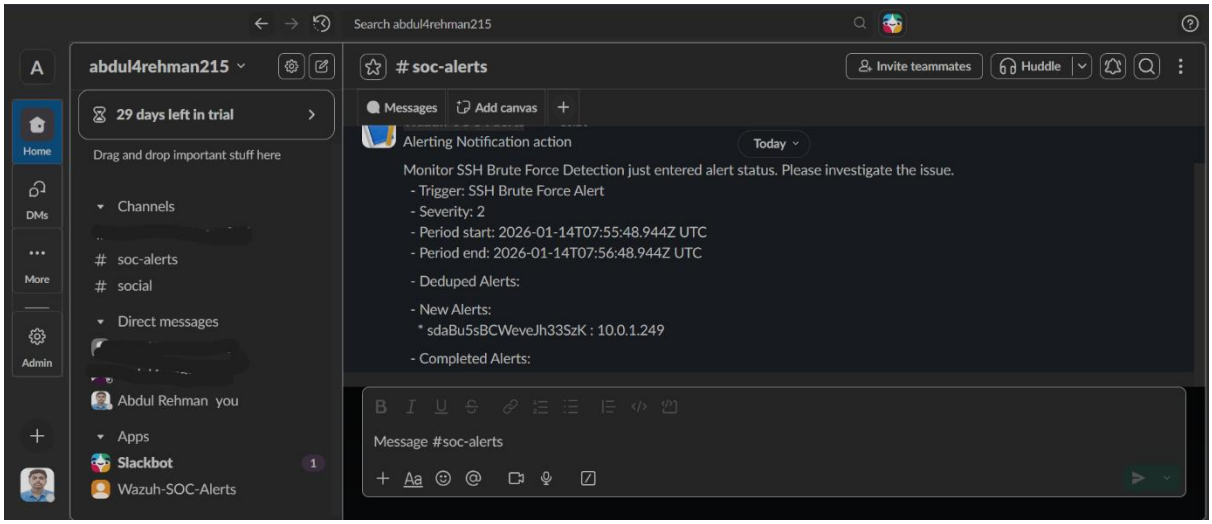






Step 14: Real-Time Alert Received in Slack

The SSH brute force alert was successfully delivered to the Slack SOC alerts channel in real time, including details such as alert name, severity, time window, and attacker IP.



❖ Outcome

This project successfully demonstrates a complete SOC detection and alerting workflow. An SSH brute force attack was simulated, detected by Wazuh, and escalated in real time via Slack. The setup reflects real-world SOC operations, including alert engineering, attack detection, and incident notification.

✓ Skills Demonstrated

- **Wazuh SIEM configuration**
- **SSH brute force attack detection**
- **SOC alert engineering**
- **Slack integration for security alerts**
- **Linux log analysis**
- **Blue team defensive security practices**